Development of Cold Chain Logistics of Dairy Products in Nanjing Using Analytical Hierarchy Process

Haibao Yu[1] Nihal Jayamaha[2] Imran Ishrat[3]

[1]Windcave

[2]School of Food & Advanced Technology, Massey University

[3]School of Business, Ara Institute of Canterbury

Ara Business Research Spotlight

08 June, 2026

Outline

Introduction

- Cold Chain Logistics (CCL) involves the management & transportation of temperature sensitive products.
- In China, CCL operations of dairy products have received significant attention in recent years.
- Nanjing, the capital city of Jiangsu Province, is selected for the study.
- Analytic Hierarchy Process (AHP): Multi-criteria decision-making (MCDM) streamlines complex decision-making.

Figure 1: Dairy Supply Chain in Nanjing

Research Question

RQ 1. What are the key factors & their significance that limit the development of CCL for dairy products in Nanjing?

RQ 2. What suggestions can be made to develop dairy CCL operations in Nanjing?

Literature Review

- CCL operations need to be well-developed, monitored & controlled to ensure food quality (Dong & Li, 2016; Liu & Shi, 2015; Wang, 2019).
- The vast distances in China pose challenges for dairy CCL operations (Wang et al., 2021; Zhang, 2017; Zhong et al., 2014).
- Government policies, infrastructure, technology, and industry practices are key factors affecting CCL in the dairy industry (Ding et al., 2019; Qian et al., 2011).
- MCDM studies are significantly used in supply chain & logistics management (Tramarico et al., 2015).
- AHP is applied in Agricultural CCL operations (Liu, 2022) and Green logistics (Niu & Xie, 2021).
- Application of AHP is used to identify dairy facility site locations in Yunnan (Liu et al., 2019), Inner Mongolia (Chen et al., 2020) and Qinzhou (Wang & Li, 2020).

Methodology

Design: Quantitative, Sample: Judgemental, Data: Questionnaire(28/58:15), Analysis: AHP

Table 1: AHP Structure

Objective	Criteria	Sub-criteria	Alternatives
How to develop cold chain logistics for dairy products in Nanjing? (A)	Government Policy (B1)	Lack of Govt. policy support (C11)	Build a complete CCL system or increase number of existing equipment and facilities for smooth CCL operations (D1)
		Inadequate regulatory system (C12)	
		Lack of technical CCL standards (C13)	
		CCL system issues (C14)	
	Infrastructure (B2)	Low level of last mile delivery (C21)	
		Insufficient number of cold storage facilities (C22)	Improve the information exchange using real-time technologies such as IoT (D2)
		Insufficient refrigerated transport vehicles (C23)	
		Low level of road transport (C24)	
	Technology (B3)	Insufficient innovation and application (C31)	
		Low level of informatisation (C32)	
		Technological backwardness (C33)	
		Imperfect CCL information systems (C34)	
	Industry Practices (B4)	Low degree of CCL specialisation (C41)	Strengthen the training of logistics professionals and develop 3PL CCL enterprises(D3)
		Insufficient number and level of 3PL enterprises (C42)	
		Insufficient awareness among the practitioners (C43)	
		Lack of professional logistics qualification (C44)	

Intensity of importance	Definition	Explanation
1	Equal importance	Two activities contribute equally to the objective
3	Moderate importance of one over another	Experience and judgment strongly favor one activity over another
5	Essential of strong importance	Experience and judgment strongly favor one activity over another
7	Very strong importance	An activity is strongly favored and its dominance demonstrated in practice
9	Extreme importance	The evidence favoring one activity over another is of the highest possible order of affirmation
2, 4, 6, 8	Intermediate values between the two adjacent judgment	When compromise is needed

Figure 2: Saaty scale

Methodology...

Judgement Matrix, $X = (a_{ij})_{m\times m}$

For criteria B, matrix $B =$

$$(\mathrm{a}_{ij})_{4\times 4} = \begin{pmatrix} a_{11} & \cdots & a_{14} \\ \vdots & \ddots & \vdots \\ a_{41} & \cdots & a_{44} \end{pmatrix}$$

For sub-criteria C, matrix $C =$

$$(\mathrm{a}_{ij})_{4\times 4} = \begin{pmatrix} a_{11} & \cdots & a_{14} \\ \vdots & \ddots & \vdots \\ a_{41} & \cdots & a_{44} \end{pmatrix}$$

Total Hierarchical Ranking

$B\&A : W^{(1)} = (\omega_1^{(1)}, \omega_2^{(1)}, \omega_3^{(1)}, \ldots, \omega_m^{(1)})^T$

$C\&B : V_j^{(2)} = (\omega_{1j}^{(2)}, \omega_{2j}^{(2)}, \omega_{3j}^{(2)}, \ldots, \omega_{mj}^{(2)})^T$

$C\&A : W^{(2)} = (V_1^{(2)}, V_2^{(2)}, V_3^{(2)}, \ldots, V_m^{(2)})^T \cdot W^{(1)}$

Hierarchical Single Ranking & Consistency Test

1) Criteria weights of X
$\bar{\omega} = (\bar{\omega}_1, \bar{\omega}_2, \bar{\omega}_3, \ldots, \bar{\omega}_m)^T$

2) Normalise vector & derive relative weights
$W = (\omega_1, \omega_2, \omega_3, \ldots, \omega_m)^T$; $\omega_i = \frac{\bar{\omega}_i}{\sum_{i=1}^{m} \bar{\omega}_i}$

3) Consistency vector $X \cdot W$
$XW = ((XW)_1, (XW)_2, (XW)_3, \ldots, (XW)_m)^T$;
$(XW)_i = \sum_{j=1}^{m} (a_{ij} \cdot \omega_j)$

4) Maximum characteristic root $\lambda_{\max} = \frac{1}{m} \sum_{i=1}^{m} \frac{(XW)_i}{\omega_i}$

5) Consistency Indicator $CI = \frac{\lambda_{\max} - m}{m-1}$ Random Indicator
$RI = 0.9$; m=4 & Consistency Ratio $CR = \frac{CI}{RI}$.
If $CR \prec 0.1$, X passes consistency test

Results

Table 2: Sub-criteria Ratios & Ranks

(a) Govt. Policy

Government Policy (B1)	Vector value	Weight ratio	Maximum characteristic root	CI	CR
Lack of govt. policy support (C11)	2.5511	52.28%	4.0609	0.0203	0.0226
Inadequate regulatory system (C12)	1.0415	21.35%			
Lack of technical CCL standards (C13)	0.8372	17.16%			
CCL system issues (C14)	0.4495	9.21%			

(b) Infrastructure

Infrastructure (B2)	Vector value	Weight ratio	Maximum characteristic root	CI	CR
Low level of last-mile delivery (C21)	0.7057	15.10%	4.0267	0.0089	0.0099
Insufficient number of cold storage facilities (C22)	0.4632	9.91%			
Insufficient number of refrigerated transport vehicles (C23)	1.8470	39.53%			
Low level of road traffic (C24)	1.6564	35.45%			

(c) Technology

Technology (B3)	Vector value	Weight ratio	Maximum characteristic root	CI	CR
Insufficient innovation (C31)	0.4897	10.27%	4.0110	0.0037	0.0041
Low level of informatization (C32)	0.6959	14.59%			
Technological backwardness (C33)	2.3171	48.59%			
Imperfect CCL information system (C34)	1.2663	26.55%			

(d) Practice

Market (B4)	Vector value	Weight ratio	Maximum characteristic root	CI	CR
Low degree of specialization in CCL (C41)	2.4144	47.19%	4.0286	0.0095	0.0106
Insufficient number and level of third-party logistics enterprises (C42)	1.7211	33.64%			
Insufficient awareness among the practitioners (C43)	0.4989	9.75%			
Lack of professional logistics qualification (C44)	0.4823	9.43%			

Results...

Table 3: Criteria Weights & Ranks

Criteria	Weight (W)	Rank	Sub-criteria	Weight	W to A	Rank
B1	15.30%	3	C11	52.28%	8.00%	**5**
			C12	21.35%	3.27%	9
			C13	17.15%	2.62%	13
			C14	9.22%	1.41%	14
B2	8.99%	4	C21	15.09%	1.36%	15
			C22	9.88%	0.89%	16
			C23	39.56%	3.56%	8
			C24	35.47%	3.19%	10
B3	47.05%	**1**	C31	10.20%	4.80%	7
			C32	14.53%	6.83%	6
			C33	48.70%	22.91%	**1**
			C34	26.57%	12.50%	**3**
B4	28.66%	2	C41	47.17%	13.52%	**2**
			C42	33.65%	9.64%	**4**
			C43	9.74%	2.79%	11
			C44	9.44%	2.70%	12

Analysis

- Technological limitations pose the greatest challenge to the advancement of CCL in Nanjing.
- Industry practices, Government policies, and Infrastructure also impede the development of CCL.
- C33, C41, C34, C42 & C11 collectively account for over 65% weight in the decision-making.
- Option 1 (D2): Improve the information exchange using Internet-of-Things (IoT) technologies
  – Expedite the integration (RFID, GPS)
- Option 2 (D3): Enhance the training of logistics professionals and develop 3P CCL
  – high specialisation, low risk
- Option 3 (D1): Build a complete CCL system
  – Increase temperature-controlled equipment (storage, transportation)
  – Strengthen food safety regulations (standards, frameworks)

Conclusion

- Research identifies the key factors and their significance affecting the development of CCL for dairy products in Nanjing.
- Technology (47.05%) $\succ$ Industry Practices (28.66%) $\succ$ Govt. Policy (15.30%) $\succ$ Infrastructure (8.99%)
- Integrate state-of-the-art technologies and strengthen 3PL capabilites before building comprehensive CCL systems.
- Future research: Examine the impact of integrating IoT devices (RFID, GPS) for real-time monitoring of all dairy CCL operations.

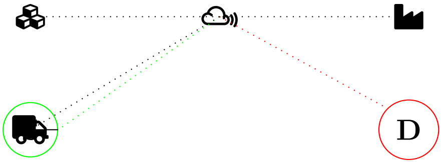

Figure 3: Upstream operations

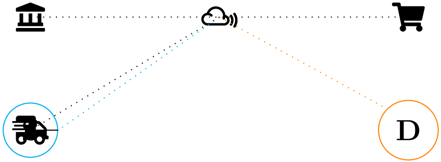

Figure 4: Downstream operations

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Questions?